

SORU 1: $S = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \right\}$, V yi gerektiririne göre V 'nin ortonormal bazını bulunuz.

$$\begin{pmatrix} 1 & 2 & 1 & 0 & 2 \\ -1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 1 & 0 & 2 \\ 0 & 3 & 2 & 1 & 3 \\ 0 & 1 & -1 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 3 & -2 & -4 \\ 0 & 0 & 5 & -2 & -6 \\ 0 & 1 & -1 & 1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 1 & 0 & 2 \\ -1 & 0 & 2 & 0 & -2 \\ 0 & 1 & -1 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & -4/5 & -4 \\ 0 & 0 & 1 & -2/5 & 6/5 \\ 0 & 1 & -1 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & -4/5 & -4 \\ 0 & 1 & -1 & 1 & 3 \\ 0 & 0 & 1 & -2/5 & -6/5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 1 & 0 & 2 \\ 0 & 2 & 3 & 0 & 0 \\ 0 & 1 & -1 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & -4/5 & -4 \\ 0 & 1 & 0 & 3/5 & 18/5 \\ 0 & 0 & 1 & -2/5 & -6/5 \end{pmatrix}$$

$\{v_1, v_2, v_3\} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$ baz
oluşturur.

$\langle v_1, v_2 \rangle = 1 \neq 0$ ortogonal değil
 $\langle v_1, v_3 \rangle = 0$, $\langle v_2, v_3 \rangle = 2 \neq 0$

$v_1 = u_1$ dersen

$u_2 = v_2 - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3/2 \\ 3/2 \\ 1 \end{bmatrix}$

$\langle v_2, u_1 \rangle = 1 \cdot (2) + (-1) \cdot (-1) + 0 \cdot (1) = 2 - 1 = 1$
 $\langle u_1, u_1 \rangle = 1 \cdot 1 + (-1) \cdot (-1) + 0 \cdot 0 = 1 + 1 = 2$

$u_3 = v_3 - \frac{\langle v_3, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 - \frac{\langle v_3, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1$

$= \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} - \frac{2}{11/2} \begin{bmatrix} 3/2 \\ 3/2 \\ 1 \end{bmatrix} - \frac{0}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5/11 \\ 5/11 \\ -15/11 \end{bmatrix}$

$\langle v_3, u_2 \rangle = 1 \cdot 3/2 + 1 \cdot 3/2 + (-1) \cdot 1 = 3 - 1 = 2$

$\langle v_3, u_1 \rangle = 1 \cdot 1 + (-1) \cdot (-1) + 0 \cdot (-1) = 1 - 1 = 0$

$\langle u_2, u_2 \rangle = \frac{3}{2} \cdot \frac{3}{2} + \frac{3}{2} \cdot \frac{3}{2} + 1 \cdot 1 = 11/2$

Ortogonal baz $\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3/2 \\ 3/2 \\ 1 \end{bmatrix}, \begin{bmatrix} 5/11 \\ 5/11 \\ -15/11 \end{bmatrix} \right\}$

ve

birim-ortonormal baz $\left\{ \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}, \begin{bmatrix} 3/\sqrt{22} \\ 3/\sqrt{22} \\ 1/\sqrt{22} \end{bmatrix}, \begin{bmatrix} 1/\sqrt{11} \\ 1/\sqrt{11} \\ -3/\sqrt{11} \end{bmatrix} \right\}$

elde edilir.

$\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{3/2}{11/2} \begin{bmatrix} 3/2 \\ 3/2 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ 1 \end{bmatrix}$

$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{6}{4}$
 $\frac{2}{\sqrt{6}} \cdot \frac{\sqrt{2}}{\sqrt{3}}$

SORU 2 $L: M_{2,1} \rightarrow P_1$, $L\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = (3a+b) + (5a+2b)x$

40P seçilmiş tanımlanmış

- a) Lineer dönüşüm old. göster 10
b) Kerkini $\text{Ker } L = ?$ bul. 10
c) L , 1-1 midir? 5
d) L , örten midir? 5
e) L^{-1} mevcut mudur? 5
 L^{-1} (ters dön) bul 10

Çözüm:

a) $\left. \begin{aligned} L(u+v) &= L(u) + L(v) \\ L(cu) &= cL(u) \end{aligned} \right\}$ old. göstermeliyiz

$$\begin{aligned} 1^o) L(u+v) &= L\left(\begin{bmatrix} a_1+a_2 \\ b_1+b_2 \end{bmatrix}\right) = [3(a_1+a_2) + (b_1+b_2)] + [5(a_1+a_2) + 2(b_1+b_2)]x \\ &= \underbrace{[3a_1 + b_1] + (5a_1 + 2b_1)x}_{L\left(\begin{bmatrix} a_1 \\ b_1 \end{bmatrix}\right)} + \underbrace{[3a_2 + b_2] + (5a_2 + 2b_2)x}_{L\left(\begin{bmatrix} a_2 \\ b_2 \end{bmatrix}\right)} \\ &= L(u) + L(v) // \end{aligned}$$

$$u = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}, v = \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}$$

$$\begin{aligned} 2^o) L(cu) &= L\left(\begin{bmatrix} ca_1 \\ cb_1 \end{bmatrix}\right) = 3(ca_1) + cb_1 + [5(ca_1) + 2(cb_1)]x \\ &= c(3a_1 + b_1) + c(5a_1 + 2b_1)x \\ &= c[3a_1 + b_1 + (5a_1 + 2b_1)x] \\ &= c \cdot L\left(\begin{bmatrix} a_1 \\ b_1 \end{bmatrix}\right) \\ &= cL(u) \end{aligned}$$

1^o) ve 2^o) sağlandığından L bir lineer dönüşümdür.

b) $\text{Ker } L$ tanımı gereği $L\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = 0_{P_1}$ den

$$3a + b + (5a + 2b)x = 0 + 0x$$

$$\left. \begin{aligned} 3a + b &= 0 \\ 5a + 2b &= 0 \end{aligned} \right\} \Rightarrow \begin{pmatrix} 3 & 1 & 0 \\ 5 & 2 & 0 \end{pmatrix} \sim \begin{pmatrix} 3 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$\text{Ker } L = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} \text{ tek elemanlı olup, boy } \text{Ker } L = 1 \text{ dir.}$$

d) Örten olması için $L(M_{2,1}) = P_1$ olmalıdır.

$$u = \begin{bmatrix} a \\ b \end{bmatrix} \in M_{2,1} \text{ için } L(u) = w \in P_1 \Rightarrow w = cx + d \text{ deriz}$$

$$\begin{aligned} L(u) &= 3a + b + (5a + 2b)x = cx + d \\ \left. \begin{aligned} 3a + b &= d \\ 5a + 2b &= c \end{aligned} \right\} \Rightarrow \begin{pmatrix} 3 & 1 & d \\ 5 & 2 & c \end{pmatrix} \sim \begin{pmatrix} 3 & 1 & d \\ -1 & 0 & c-2d \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & 3c-6d+c \\ 1 & 0 & 2d-c \end{pmatrix} \\ a &= 2d-c \end{aligned}$$

rank=2
tek çözüm

$\Rightarrow L$ örtendir.

veya

$$\underbrace{\text{boy Ker } L}_1 + \underbrace{\text{boy } L(M_{21})}_2 = \underbrace{\text{boy } (M_{12})}_2 \quad 5$$

sağlanır.

O halde L önterdir.

c) L 1-1 midir?

$$\text{Ker } L = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} \quad \text{Tek elementli olup, } L(M_{21}) = O_{P_1} \text{ dir.}$$

$$\text{yani } L\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = O_{P_1}$$

$$3a+b + (5a+2b)x = 0+0x$$

$$\left. \begin{array}{l} 3a+b=0 \\ 5a+2b=0 \end{array} \right\} \Rightarrow \begin{pmatrix} 3 & 1 & 0 \\ 5 & 2 & 0 \end{pmatrix} \sim \begin{pmatrix} 3 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$a=0, b=0$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{sağlanır.}$$

$$\text{Yani } \text{Ker } L = \{ O_{M_{12}} \} \text{ dir.}$$

$\Rightarrow L$, 1-1 dir.

e) L^{-1} mevcut mudur? $L^{-1} = ?$

(2)

L , 1-1 ve örten old. L^{-1} bulunur.

$$L^{-1}\left(\underbrace{w}_{\in P_1}\right) = u, \quad w = cx+d, \quad u = \begin{bmatrix} a \\ b \end{bmatrix} \in M_{21}, w \in P_1$$

$$L^{-1}(cx+d) = c L^{-1}(x) + d L^{-1}(1)$$

$$L^{-1}(1) = L^{-1}(1+0x) = u \quad 2$$

$$L(u) = L\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = 3a+b + (5a+2b)x = w = cx+d$$

$$\left. \begin{array}{l} 3a+b=d \\ 5a+2b=c \end{array} \right\} \Rightarrow \begin{array}{l} a=2d-c \\ b=3c-5d \end{array} \text{ bulunmuştur.}$$

$$L^{-1}\left(\underbrace{1+0x}_{c=0, d=1}\right) = L^{-1}(1) \text{ den } \underline{a=2, b=-5} \Rightarrow \begin{bmatrix} 2 \\ -5 \end{bmatrix}$$

$$L^{-1}\left(\underbrace{0+1 \cdot x}_{c=1, d=0}\right) = L^{-1}(x) \text{ den } \underline{a=-1, b=3} \Rightarrow \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$L^{-1}(cx+d) = c L^{-1}(x) + d L^{-1}(1)$$

$$= c \begin{bmatrix} -1 \\ 3 \end{bmatrix} + d \begin{bmatrix} 2 \\ -5 \end{bmatrix} \Rightarrow \begin{bmatrix} -c+2d \\ 3c-5d \end{bmatrix} = L^{-1}(cx+d)$$

bulunur.

de yerine yazılır